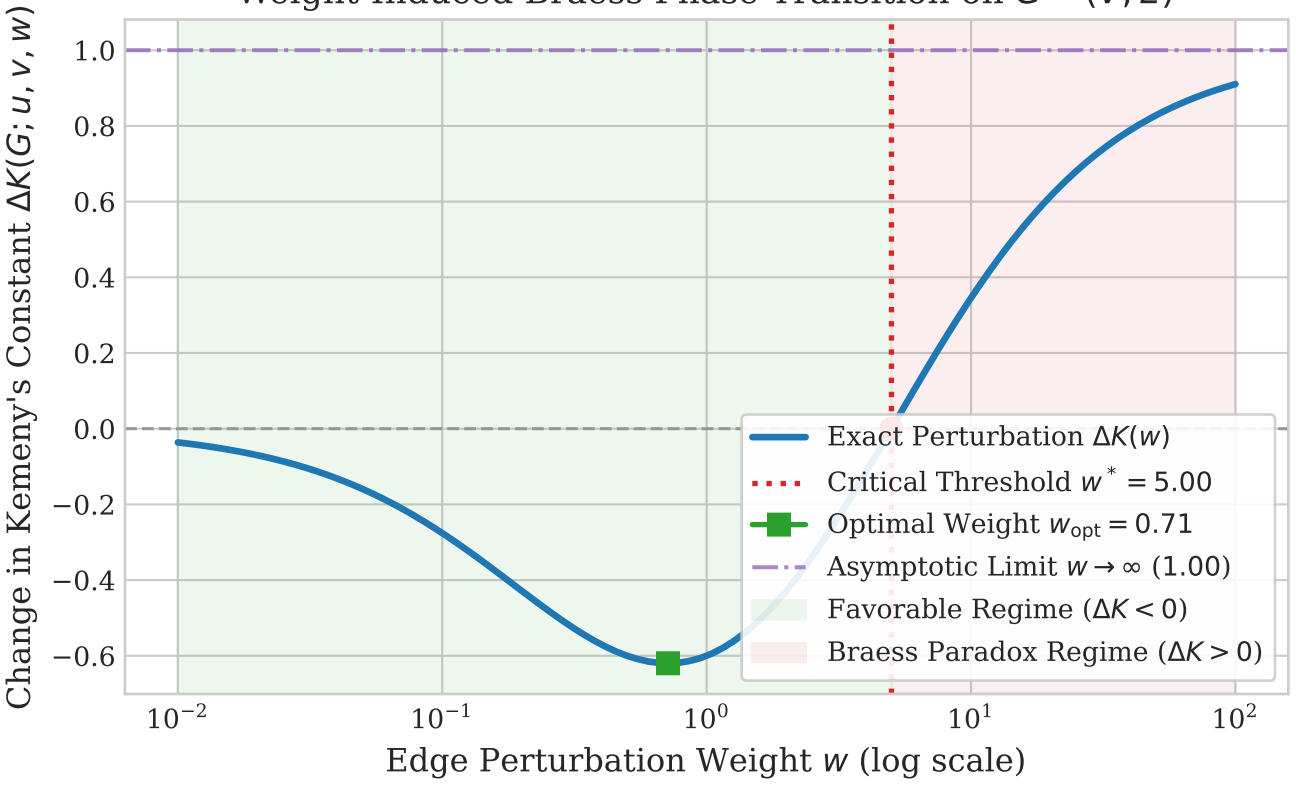
Weight-Induced Braess Phase Transition on $G = (V, E)$
Change in Kemeny's Constant $\Delta K(G; u, v, w)$
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Edge Perturbation Weight w (log scale)
Exact Perturbation $\Delta K(w)$
Critical Threshold $w^* = 5.00$
Optimal Weight $w_{\text{opt}} = 0.71$
Asymptotic Limit $w \to \infty$ (1.00)
Favorable Regime ($\Delta K < 0$)
Braess Paradox Regime ($\Delta K > 0$)